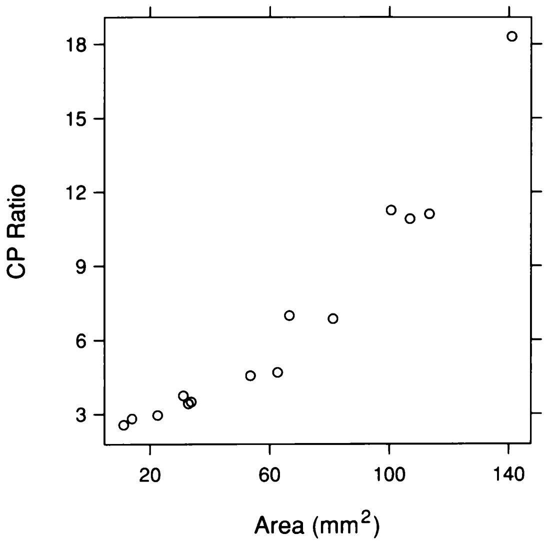


3.1 A scatterplot displays bivariate data: measurements of retinal area and CP ratio for 14 cat fetuses.

3 Bivariate Data

Figure 3.1 is a scatterplot of two variables from a study in animal physiology [60]. For species with highly developed visual systems, such as cats and man, the distribution of ganglion cells across the surface of the retina is not uniform. For example, cats at birth have a much greater density of cells in the central portion of the retina than on the periphery. But in the early stages of fetal development, the distribution of ganglion cells is uniform. The nonuniformity develops in later stages. Figure 3.1 is a scatterplot of data for 14 cat fetuses ranging in age from 35 to 62 days of gestation. The vertical scale is the ratio of the central ganglion cell density to the peripheral density. The horizontal scale is the retinal area; this variable is nearly monotonically increasing with age, so age tends to increase from left to right along the scale. The scatterplot shows that the ratio increases as the retinal area increases.

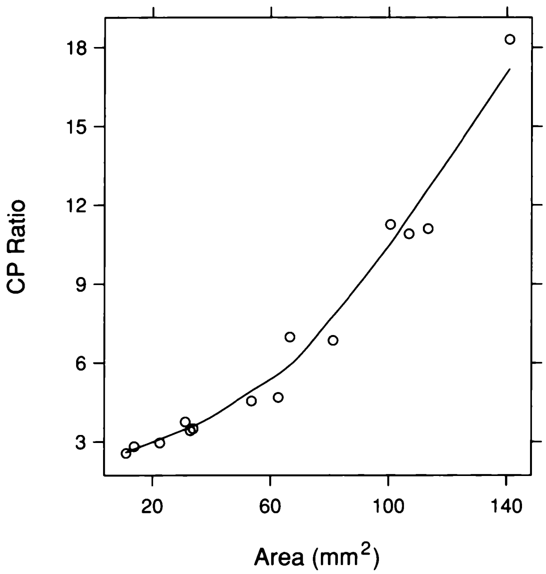
The ratios and areas in Figure 3.1 are *bivariate data*: paired measurements of two quantitative variables. The goal in this case is to study how the ratios depend on the areas. Ratio is the *response* and area is the *factor*. In other examples of this chapter, neither variable is a factor or response, and the goal is to study the distribution of the observations in the plane and to determine how the variables are related.

3.1 Smooth Curves and Banking

The scatterplot is an excellent first exploratory graph to study the dependence of a response on a factor. An important second exploratory graph adds a smooth curve to the scatterplot to help us better perceive the pattern of the dependence.

Loess Curves

Figure 3.2 adds a *loess* curve to the ganglion data; loess is a fitting tool that will be described in the next section [19, 22, 23]. The fitted curve has a convex pattern, which means that the rate of change of CP ratio with area increases as area increases. In this example, the existence of the curvature is apparent from the unadorned scatterplot in Figure 3.1, but the graphing of the curve provides a more incisive description. In examples coming later, even the broad nature of the curvature is not revealed until a curve is added.



3.2 A loess curve has been added to the scatterplot of the ganglion data. The aspect ratio of the graph has been chosen to bank the curve to 45°.

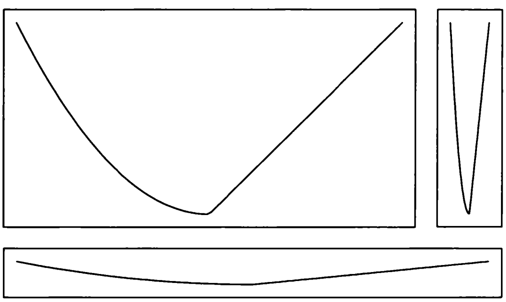
Visual Perception: Banking

The *data rectangle* of a graph just encloses the data on the graph. In Figure 3.2, the upper right corner of the data rectangle is at the data point in the upper right corner, and the lower left corner of the rectangle is at the data point in the lower left corner. The *aspect ratio* of a graph is the height of the data rectangle divided by the width. In Figure 3.2, the aspect ratio is 1.08.

The fitted curve in Figure 3.2 is made up of a collection of connected line segments. First, the fit was evaluated at 100 equally spaced values of the factor, which results in 100 points in the plane; then the curve was drawn by connecting the 100 points by line segments.

Our perception of a curve is based on judgment of the relative orientations of the line segments that make up the curve. Suppose we measure orientation in degrees. A line segment with slope 1 has an orientation of 45°, and a line segment with slope -1 has an orientation of -45°. Typically, the judgments of a curve are optimized when the absolute values of the orientations are centered on 45°, that is, when the location of the distribution of absolute orientations is 45° [21, 25]. Chapter 4 contains an analysis of the data from the perceptual experiment that led to the discovery of this 45° principle.

It is the aspect ratio that controls the orientations of line segments on a graph. This is illustrated in Figure 3.3, which draws the same curve with three aspect ratios: 0.5 in the upper left panel, 0.05 in the lower panel, and 5 in the upper right panel. The absolute orientations are centered on 45° in the upper left, on 5.7° in the lower, and on 84° in the upper right. The upper left panel provides the best perception of the properties of the curve; the left half of the curve is convex and the right half is linear. Choosing the aspect ratio of a graph to enhance the perception of the orientations of line segments is *banking*, a display method whose name suggests the banking of a road to affect its slope. Choosing the aspect ratio to center the absolute orientations on 45° is *banking to 45°*. In Figure 3.2, the aspect ratio has been chosen by banking the segments of the loess curve to 45°. At the end of this section, a method will be given for finding the aspect ratio that banks a collection of segments to 45°.



3.3 The same curve is graphed with three different aspect ratios: 0.5 in the upper left panel, 5 in the upper right panel, and 0.05 in the bottom panel. The aspect ratio of 0.5 banks the curve to 45°.

Banking is a vital aspect of visualizing curves, and will be widely invoked in this and later chapters. It is not always practical to bank to 45° because the aspect ratio that produces it can be too small or too large for the resolution of the physical display device. In some cases, as we will see later in this chapter, a display can be redesigned to enable the optimal value. In other cases we will simply take the aspect ratio to be as close to the optimal value as possible.

For the Record: A Method for Banking to 45°

Let v be the length in physical units of a vertical side of the data rectangle of a graph. In Figure 3.2, the scatterplot of the ganglion data with the loess curve, the value of v is 5.5 cm. Let $\ddot{v}$ be the length of the vertical side in data units. In Figure 3.2, the value of $\ddot{v}$ is 15.7 units of CP ratio, or 15.7 cpr. Similarly, let h be the length in physical units of a horizontal side of the data rectangle, and let $\ddot{h}$ be the length in data units.

Consider the length in data units of any interval on the measurement scale of a variable. For example, suppose for the CP ratio scale that the length is 5 cpr. Then $v/\ddot{v}$ is the conversion factor that takes the length in the units of the data and changes it to a length in physical units on the graph. For example, for Figure 3.2, 5 cpr becomes

$$5 \text{ cpr } \frac{5.5 \text{ cm}}{15.7 \text{ cpr}} = 1.8 \text{ cm} .$$

Similarly, $h/\ddot{h}$ is the conversion factor for the horizontal scale.

The aspect ratio of the data on the graph is

$$a(h, v) = v/h .$$

Suppose the units of the data are fixed so that $\ddot{v}$ and $\ddot{h}$ are fixed values. The values of v and h are under our control in graphing the data. The aspect ratio is determined by our choice of v and h, which is why in the notation we show the dependence of a on v and h.

Consider a collection of n line segments inside the data region. Let $\ddot{v}_i$ be the change in data units of the ith line segment along the vertical scale, and let $v_i(v)$ be the change in physical units when the vertical length of the data rectangle is v. Define $\ddot{h}_i$ and $h_i(h)$ similarly. Let

$$\bar{v}_i = \ddot{v}_i/\ddot{v}$$

and

$$\bar{h}_i = \ddot{h}_i/\ddot{h} .$$

The orientation of the ith segment is

$$\theta_i(h, v) = \arctan\left(\frac{v_i(v)}{h_i(v)}\right) = \arctan\left(a(h, v)\bar{v}_i/\bar{h}_i\right) .$$

The physical length of the ith segment is

$$\ell_i(h, v) = \sqrt{h_i^2(h) + v_i^2(v)} = h\sqrt{\bar{h}_i^2 + a^2(h, v)\bar{v}_i^2} .$$

One method for banking the n segments to 45° is to choose $a(h, v)$ so that the mean of the absolute orientations weighted by the line segment lengths is 45°. Thus

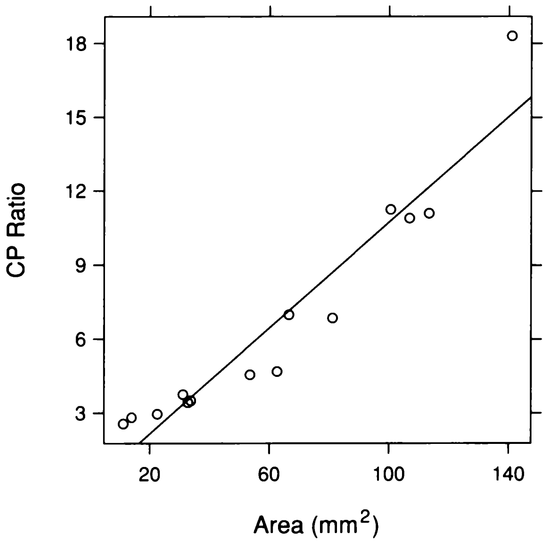
$$\frac{\sum_{i=1}^{n} \theta_i(h, v)\ell_i(h, v)}{\sum_{i=1}^{n} \ell_i(h, v)} = \frac{\sum_{i=1}^{n} \arctan\left(a(h, v)\bar{v}_i/\bar{h}_i\right) \sqrt{\bar{h}_i^2 + a^2(h, v)\bar{v}_i^2}}{\sum_{i=1}^{n} \sqrt{\bar{h}_i^2 + a^2(h, v)\bar{v}_i^2}}$$

is equal to 45°. Notice that the right side of this formula depends on v and h only through $a(h, v)$. As is intuitively clear, if we multiply v and h by the same factor, the orientations of the segments do not change. Only their ratio matters. Thus it is the aspect ratio that controls banking.

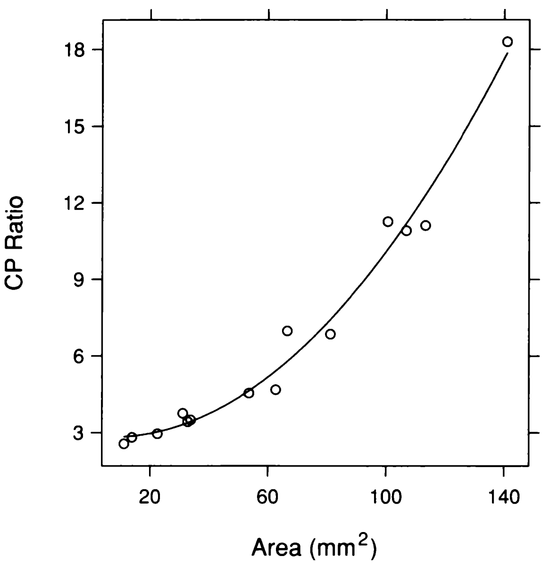
There is no closed-form solution for the aspect ratio that makes the above weighted mean absolute orientation equal to 45°. The value of a needs to be found by iterative approximation. But the approximation can be fast because the weighted mean is a monotone function of a.

3.2 *Fitting: Parametric and Loess*

Figure 3.2 has shown clearly that the underlying pattern in the ganglion data is convex. Astonishingly, the three experimenters who gathered and analyzed the data, fitted a line. Figure 3.4 shows the fit. As expected, there is *lack of fit*. Figure 3.5 shows the fit of a quadratic. The fitted function now describes the underlying pattern in the data. The lack of fit has been eliminated.



3.4 The least-squares line has been added to the scatterplot of the ganglion data.



3.5 The least-squares fit of a quadratic polynomial has been added to the scatterplot of the ganglion data. The curve is banked to 45°.

Parametric Families of Functions

Fitting a line to data is an example of *parametric fitting*. Lines, or linear polynomials,

$$y = a + bx ,$$

are a parametric family of functions that depend on two parameters, a and b. Fitting the linear family to data means selecting, or estimating, a and b so that the resulting line best describes the underlying dependence of y on x. Quadratic polynomials,

$$y = a + bx + cx^2 ,$$

are a parametric family with three parameters: a, b, and c.

The classical method for fitting parametric families is *least-squares*. The parameters are estimated by finding the member of the family that minimizes the sum of the squares of the vertical deviations of the data from the fitted function. For example, for linear polynomials, the least-squares estimates of a and b are the values that minimize

$$\sum_{i=1}^{n}(y_i - a - bx_i)^2 ,$$

where the x_i are the measurements of the factor, and the y_i are the measurements of the response. Least-squares was used for the linear and quadratic fits to the ganglion data. The linear fit is

$$y = 0.0140 + 0.107x ,$$

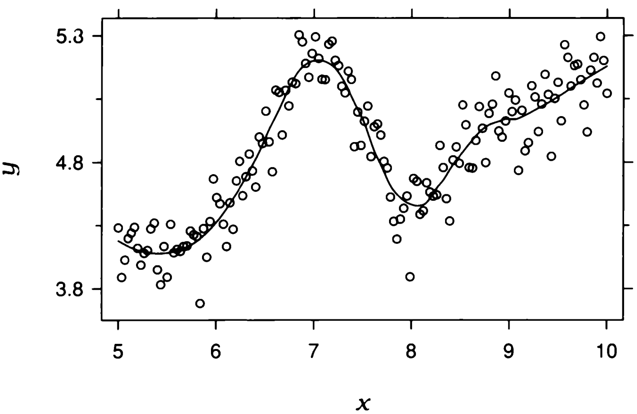
and the quadratic fit is

$$y = 2.87 - 0.0120x + 0.000839x^2 .$$

Least-squares is a good method to use provided the data have certain properties; checking to see if the data have these properties will be taken up in the next section.

Flexible Fitting

Parametric fitting is very useful but not nearly enough. When the underlying pattern in a set of data is simple — for example, the quadratic pattern in the ganglion data — we can expect to find a parametric family that will provide a good fit. But nature is frequently not so obliging. The patterns in many bivariate data sets are too complex to be described by a simple parametric family. For visualization, we need a method of fitting that will handle these complex cases. Loess fitting does just this. Figure 3.6 shows an example. The loess curve on the graph follows a pattern that defies fitting by a simple parametric function.



3.6 The data are fitted by loess. The curve is banked to 45°.

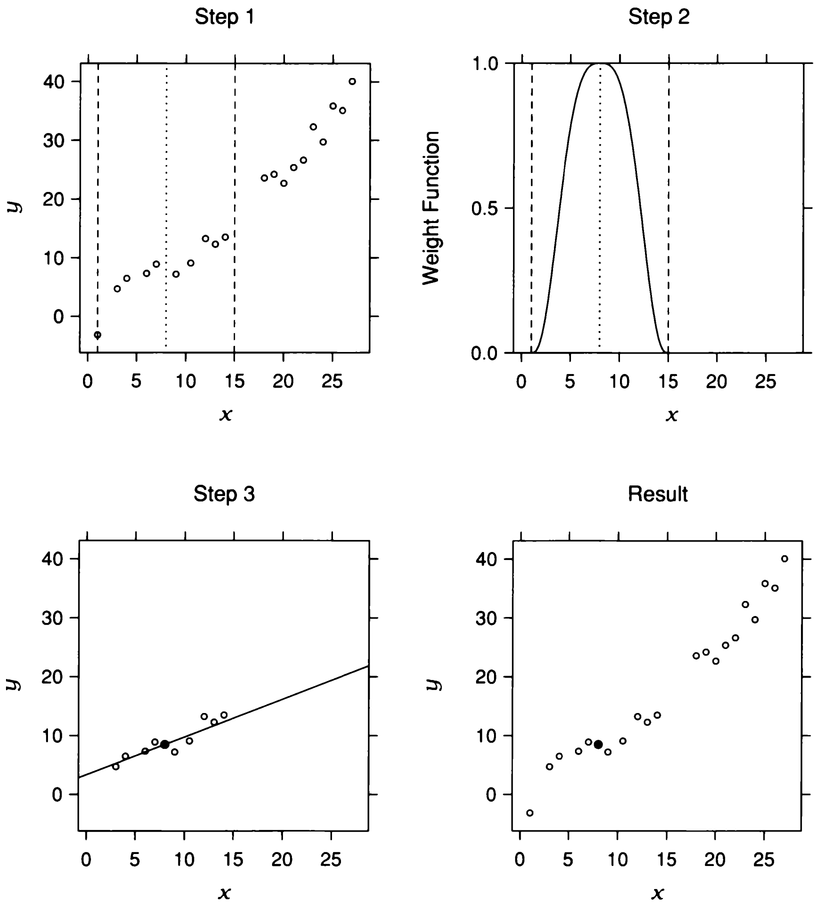
The name *loess* — pronounced (*lō´ is*) and from the German *löss* — is short for *local regression*. The name also has semantic substance. In Chapter 4, loess is used to fit surfaces to data in three dimensions, and geologists and hikers know well that a loess is a surface of sorts — a deposit of silt, or fine clay, typically found in valleys. Loess is just one of many curve-fitting methods that are often referred to as nonparametric regression [49]. But loess has some highly desirable statistical properties [36, 48], is easy to compute [23], and, as we will see, is easy to use.

A Graphical Look at How Loess Works

Two parameters need to be chosen to fit a loess curve. The first parameter, α, is a smoothing parameter; it can be any positive number but typical values are $1/4$ to 1. As α increases, the curve becomes smoother. The second parameter, λ, is the degree of certain polynomials that are fitted by the method; λ can be 1 or 2. Guidance on the choices of α and λ will be given later.

Figure 3.7 shows how the loess fit, $\widehat{g}(x)$, is computed at $x = 8$ for data that are graphed by the circles in the upper left panel. The values of the loess parameters are $\lambda = 1$ and $\alpha = 0.5$. The number of observations, $n = 20$, is multiplied by α, which gives the value 10. A vertical strip, depicted by the dashed vertical lines in the upper left panel of Figure 3.7, is defined by centering the strip on x and putting one boundary at the 10th closest x_i to x.

The observations $(x_i,\ y_i)$ are assigned *neighborhood weights*, $w_i(x)$, using the weight function shown in the upper right panel. The function has a maximum at 8, decreases as we move away from this value in either direction, and becomes zero at the boundaries of the strip. The observations whose x_i lie closest to x receive the largest weight, and observations further away receive less.



3.7 The display shows how the loess fit is computed at $x = 8$.

A line is fitted to the data using weighted least-squares with weight $w_i(x)$ at (x_i, y_i). The fit is shown in the lower left panel. The $w_i(x)$ determine the influence that each $(x_i,\ y_i)$ has on the fitting of the line. The influence decreases as x_i increases in distance from x, and finally becomes zero. The loess fit, $\widehat{g}(x)$, is the value of the line at x, depicted by the filled circle.

Thus the computations illustrated in Figure 3.7 result in one value of the loess fit, $\widehat{g}(8)$. This value is shown by the filled circle in the lower right panel.

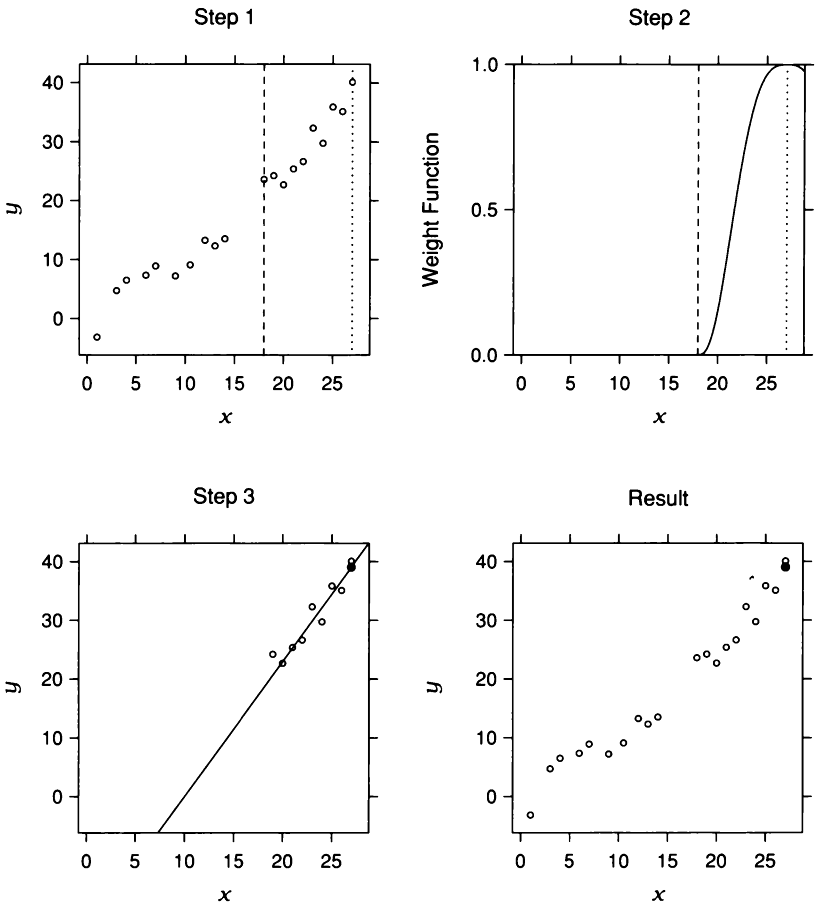
The same operations can be carried out for any other value of x. Figure 3.8 shows them at $x = 27$, which happens to be the value of the largest x_i. The right boundary of the strip does not appear in the top panels because it is beyond the right extreme of the horizontal scale line.

Since $\lambda = 1$, a linear polynomial has been fitted in Figures 3.7 and 3.8. This is *locally linear fitting*. If $\lambda = 2$, a quadratic polynomial is used, which is *locally quadratic fitting*.

A loess fit is displayed by evaluating it at a grid of equally spaced values from the minimum value of the x_i to the maximum, and then connecting these evaluated curve points by line segments. The number of evaluations is typically in the range 50 to 200.

The Two Loess Parameters

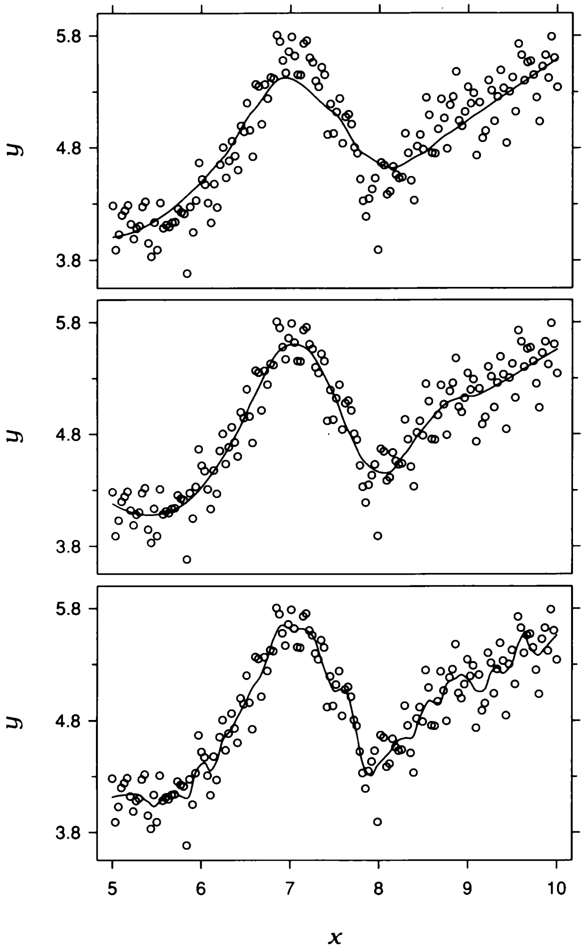
In any specific application of loess, the choice of the two parameters α and λ must be based on a combination of judgment and of trial and error. There is no substitute for the latter. As the chapter progresses, many examples will illustrate how residual plots allow us to judge a particular choice. But the following description of the influence of changing the parameters, together with experience in using loess, can help to provide educated first guesses.



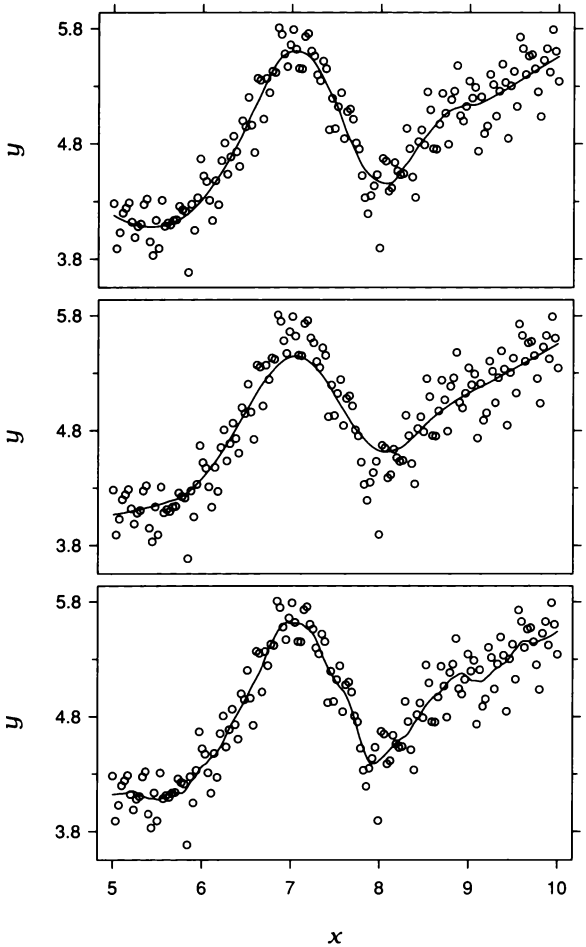
3.8 The display shows how the loess fit is computed at $x = 27$.

As α increases, a loess fit becomes smoother. This is illustrated in Figure 3.9. In all panels, $\lambda = 2$, so the fitting is locally quadratic. In the top panel, $\alpha = 0.6$. The fitted function is very smooth, but there is *lack of fit*; both the peak and the valley are missed because α is too large. In the middle panel, α has been reduced to 0.3. Now the curve is less smooth, but it follows the pattern of the data. In the bottom panel, α has been reduced still further to 0.1. The underlying pattern is tracked, but there is *surplus of fit*; the local wiggles do not appear to be supported by the data. The goal in choosing α is to produce a fit that is as smooth as possible without unduly distorting the underlying pattern in the data. In this example, $\alpha = 0.3$ is a good choice that follows the pattern without undue wiggles.

If the underlying pattern of the data has gentle curvature with no local maxima and minima, then locally linear fitting is usually sufficient. But if there are local maxima or minima, then locally quadratic fitting typically does a better job of following the pattern of the data and maintaining local smoothness. This is illustrated in Figure 3.10. The top panel shows the fit from the middle panel of Figure 3.9, which has $\lambda = 2$ and $\alpha = 0.3$. The substantial curvature, in particular the peak and valley, are adequately tracked by the curve. In the middle panel of Figure 3.10, λ has been changed to 1, with α remaining at 0.3. The fit does poorly. Because of the curvature in the data, locally linear fitting is not able to accommodate the peak and valley; the curve lies below the peak and above the valley. In the bottom panel, $\lambda = 1$, and α has been reduced to 0.1 in an attempt to follow the pattern. The curve tracks the data, but it has surplus of fit because α is small. But if we increase α by more than just a very small amount, still using locally linear fitting, the curve becomes smoother but no longer tracks the data. Locally linear fitting is not capable of providing an adequate fit in this example.



3.9 The three loess curves have three different values of the smoothing parameter, α. From the bottom panel to the top the values are 0.1, 0.3, and 0.6. The value of λ is 2.



3.10 Three loess fits are shown. From the bottom panel to the top the two parameters, α and λ, are the following: 0.1 and 1; 0.3 and 1; 0.3 and 2.

For the Record: The Mathematical Details of Loess

The following describes in detail the loess fit, $\widehat{g}(x)$, at x. Suppose first that $\alpha \leq 1$. Let q be αn truncated to an integer. We will assume that α is large enough so that q is at least 1, although in most applications, q will be much larger than 1.

Let $\Delta_i(x) = |x_i - x|$ be the distance from x to the x_i, and let $\Delta_{(i)}(x)$ be these distances ordered from smallest to largest. Let $T(u)$ be the *tricube weight function*:

$$T(u) = \begin{cases} (1 - |u|^3)^3 & \text{for } |u| < 1 \\ 0 & \text{otherwise} . \end{cases}$$

Then the neighborhood weight given to the observation (x_i, y_i) for the fit at x is

$$w_i(x) = T\left(\frac{\Delta_i(x)}{\Delta_{(q)}(x)}\right) .$$

For x_i such that $\Delta_i(x) < \Delta_{(q)}(x)$, the weights are positive and decrease as $\Delta_i(x)$ increases. For $\Delta_i(x) \geq \Delta_{(q)}(x)$, the weights are zero. If $\alpha > 1$, the $w_i(x)$ are defined in the same manner, but $\Delta_{(q)}(x)$ is replaced by $\Delta_{(n)}(x)\alpha$.

If $\lambda = 1$, a line is fitted to the data using weighted least-squares with weight $w_i(x)$ at (x_i, y_i); values of a and b are found that minimize

$$\sum_{i=1}^{n} w_i(x)\,(y_i - a - bx_i)^2 .$$

Let $\hat{a}$ and $\hat{b}$ be the minimizing values, then the fit at x is

$$\hat{g}(x) = \hat{a} + \hat{b}x .$$

If $\lambda = 2$, a quadratic is fitted to the data using weighted least-squares; values of a, b, and c are found that minimize

$$\sum_{i=1}^{n} w_i(x)\,(y_i - a - bx_i - cx_i^2)^2 .$$

Let $\hat{a}$, $\hat{b}$, and $\hat{c}$ be the minimizing values, then the fit at x is

$$\hat{g}(x) = \hat{a} + \hat{b}x + \hat{c}x^2 .$$

3.3 Visualizing Residuals

In Chapter 2, one method for fitting univariate data was to estimate the locations of the distributions. An example is the singer heights; let h_{pi} be the ith height for the pth voice part. Locations were estimated by the voice-part means, $\bar{h}_p$. Each height has a fitted value,

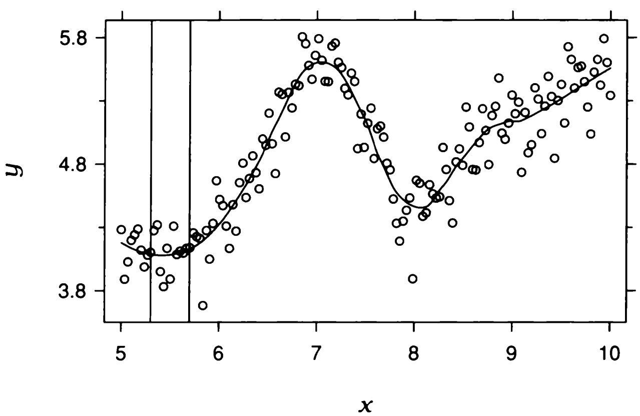
$$\widehat{h}_{pi} = \bar{h}_p \ ,$$

and a residual,

$$\widehat{\varepsilon}_{pi} = h_{pi} - \widehat{h}_{pi} \ .$$

The fitted values account for the variation in the heights attributable to the voice-part variable through the fitting process. The residuals are the remaining variation in the data.

The fitting in this chapter has so far followed a parallel course: estimating location. Whether we use parametric fitting or loess, the result is a function, $y = \widehat{g}(x)$, that describes the location of the measurements of the response, y, when the value of the factor is x. This is illustrated by the loess fit, $\widehat{g}(x)$, in Figure 3.11. The vertical strip contains observations whose x values are nearly the same. The loess fit in the strip, which is nearly constant, describes the location of the



3.11 The display illustrates the purpose of curve fitting — describing the location of the distribution of the response given the factor.

distribution of the y values of observations in the strip. Furthermore, for each observation, (x_i, y_i), there is a fitted value

$$\widehat{y}_i = \widehat{g}(x_i) \ ,$$

and a residual

$$\widehat{\varepsilon}_i = y_i - \widehat{y}_i \ .$$

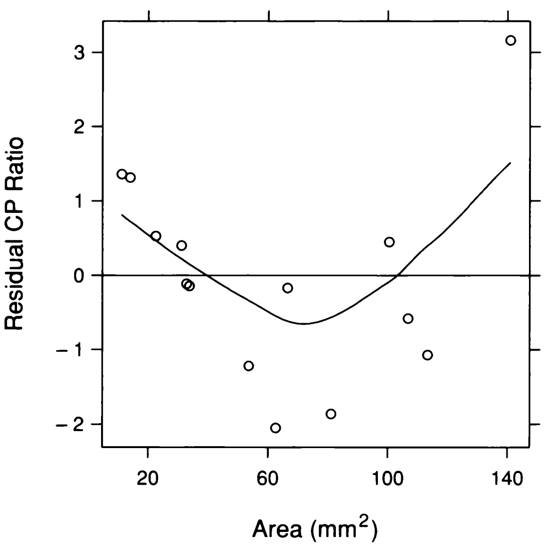
The $\widehat{y}_i$ are the variation in the response, y, attributable to the factor, x, through the fitting process. The $\widehat{\varepsilon}_i$ are the remaining variation. Thus the y_i are decomposed into fitted values and residuals,

$$y_i = \widehat{y}_i + \widehat{\varepsilon}_i \ .$$

And as with univariate data, an important part of visualizing bivariate data is the graphing of residuals from fits.

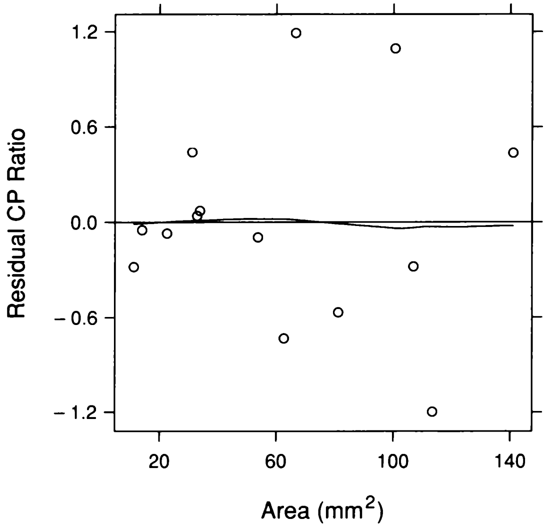
Residual Dependence Plots

If a fitted function suffers from lack of fit, the aspect of the underlying pattern not accounted for by the fit comes out in the residuals. A line did not fit the ganglion data. Figure 3.12 graphs the residuals against area together with a loess curve. This *residual dependence plot* shows a remaining dependence of the residuals on area. The location of the distribution of the residuals still varies with area.



3.12 On this residual dependence plot, the residuals from the linear fit to CP ratio are graphed against area. The parameters of the loess curve on the plot are $\alpha = 1$ and $\lambda = 1$.

If a fit is adequate, the location of the residual distribution at each x should be zero. A quadratic appeared to adequately fit the ganglion data. Figure 3.13 is a residual dependence plot. The location of the residual distribution no longer depends on area.



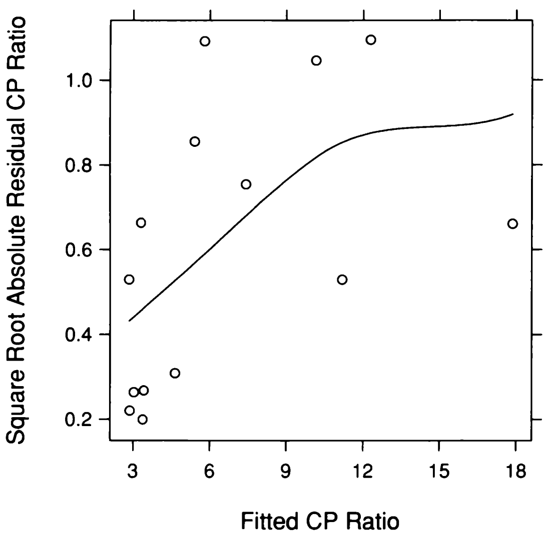
3.13 On this residual dependence plot, the residuals from the quadratic fit to CP ratio are graphed against area. The parameters of the loess curve on the plot are $\alpha = 1$ and $\lambda = 1$.

A residual dependence plot is an excellent diagnostic for detecting lack of fit when it exists. Of course, for the linear fit to the ganglion data, lack of fit is so blatant that it is obvious just from Figure 3.4, the scatterplot of the data with the fit superposed. But in other cases, lack of fit can be more subtle, and we must rely on the residual dependence plot to make a reliable judgment.

S-L Plots

The quadratic fit to the ganglion data has resulted in residuals whose location does not depend on area. Still, the residuals might not be homogeneous. Aspects of the residual distribution other than location might depend on area. As with univariate data, one type of nonhomogeniety is monotone spread: the spread of the residuals increases or decreases as the fitted values increase. And as with univariate data, we can check for monotone spread by an s-l plot.

Figure 3.14 is an s-l plot for the quadratic fit to the ganglion data. The square roots of the absolute residuals, $\sqrt{|\hat{\varepsilon}_i|}$, are graphed against the fitted values, $\hat{y}_i$, and a loess curve is added. There is monotone spread. Thus the residuals are not homogeneous, which makes the characterization of the dependence of the response on the factor more complicated. When the residuals are homogeneous, the residual variation at any x can be described by the distribution of all of the residuals. In other words, the power of pooling, discussed in Chapter 2, can be brought to bear. But if the residual variation changes, we must take the change into account in characterizing the variation.



3.14 On this s-l plot, the square root absolute residuals from the quadratic fit to CP ratio are graphed against the fitted values to check for monotone spread. The parameters of the loess curve are $\alpha = 2$ and $\lambda = 1$.